200mA, 30V Schottky Barrier Diode

FEATURES

- AEC-Q101 qualified
- Fast switching speed
- Low forward voltage drop
- Surface mound device type
- Moisture sensitivity level: level 1, per J-STD-020
- RoHS Compliant
- Halogen-free

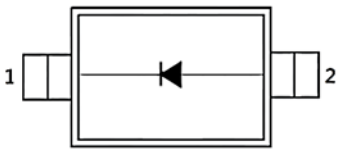
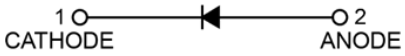
KEY PARAMETERS		
PARAMETER	**VALUE**	**UNIT**
I_F	200	mA
V_{RRM}	30	V
I_{FSM}	4	A
$T_{J\ MAX}$	125	°C
V_F at I_F=200mA	1	V
Configuration	Single die	

APPLICATIONS

- For general purpose applications

MECHANICAL DATA

- Case: SOD-323
- Molding compound meets UL 94V-0 flammability rating
- Terminal: Matte tin plated leads, solderable per J-STD-002
- Meet JESD 201 class 2 whisker test
- Polarity: Indicated by cathode band
- Weight: 4.5mg (approximately)

PACKAGE: SOD-323	PIN CONFIGURATION	CIRCUIT DIAGRAM
		

ABSOLUTE MAXIMUM RATINGS (T_A = 25°C unless otherwise noted)

PARAMETER		SYMBOL	VALUE	UNIT
Power dissipation(1)		P_D	200	mW
Repetitive peak reverse voltage		V_{RRM}	30	V
Forward current		I_F	200	mA
Repetitive peak forward current	t < 1.0s	I_{FRM}	500	mA
Non-repetitive peak forward surge current	t = 8.3ms	I_{FSM}	4	A
Junction temperature		T_J	-55 to +125	°C
Storage temperature		T_{STG}	-55 to +150	°C

Note:

1. Device mounted on an FR4 Printed-Circuit Board (PCB), single-sided copper, tin-plated and standard footprint

THERMAL PERFORMANCE

PARAMETER	SYMBOL	TYP	UNIT
Junction-to-ambient thermal resistance(1)	$R_{\Theta JA}$	500	°C/W

Thermal Performance Note:

1. Device mounted on an FR4 Printed-Circuit Board (PCB), single-sided copper, tin-plated and standard footprint

ELECTRICAL SPECIFICATIONS (T_A = 25°C unless otherwise noted)

PARAMETER	CONDITIONS		SYMBOL	MIN	TYP	MAX	UNIT
Forward voltage(1)	BAT42WSH	I_F = 10mA	V_F	-	-	0.40	V
		I_F = 50mA		-	-	0.65	
		I_F = 200mA		-	-	1.00	
	BAT43WSH	I_F = 2mA		-	-	0.33	
		I_F = 15mA		-	-	0.45	
		I_F = 200mA		-	-	1.00	
Reverse breakdown voltage	I_R = 100μA		V_{BR}	30	-	-	V
Reverse current(2)	V_R = 25V		I_R	-	-	500	nA
	V_R = 25V, T_J = 100°C			-	-	100	μA
Junction capacitance	f = 1MHz, V_R = 1V		C_J	-	-	10	pF
Reverse recovery time	I_F = I_R = 10mA, I_{RR} = 1mA, R_L = 100Ω		t_{rr}	-	-	5	ns

Notes:

1. Pulse test with PW=0.3 ms
2. Pulse test with PW=30 ms

ORDERING INFORMATION

ORDERING CODE	PACKAGE	PACKING	DEVICE MARKING
BAT42WSH RRG	SOD-323	3,000 / 7” Tape & Reel	S7
BAT43WSH RRG	SOD-323	3,000 / 7” Tape & Reel	S8

CHARACTERISTICS CURVES

(T_A = 25°C unless otherwise noted)

Fig.1 Power Dissipation Curve

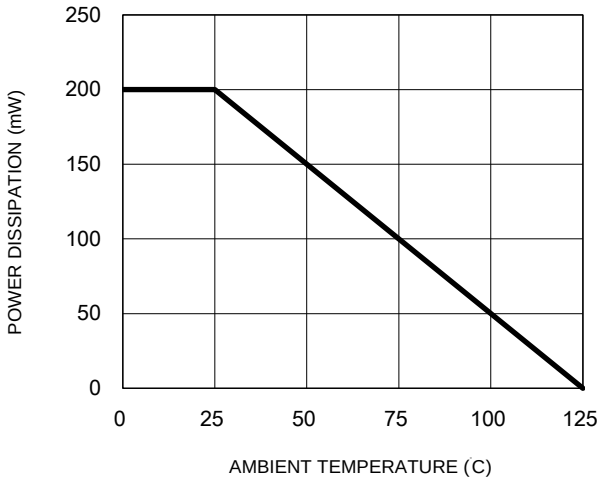


Fig.2 Typical Junction Capacitance

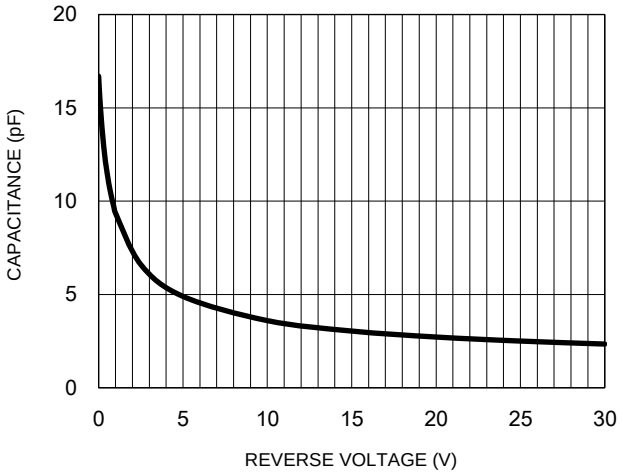


Fig.3 Typical Reverse Characteristics

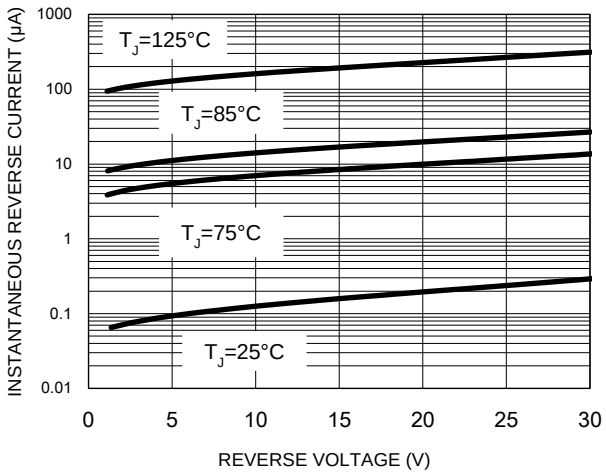
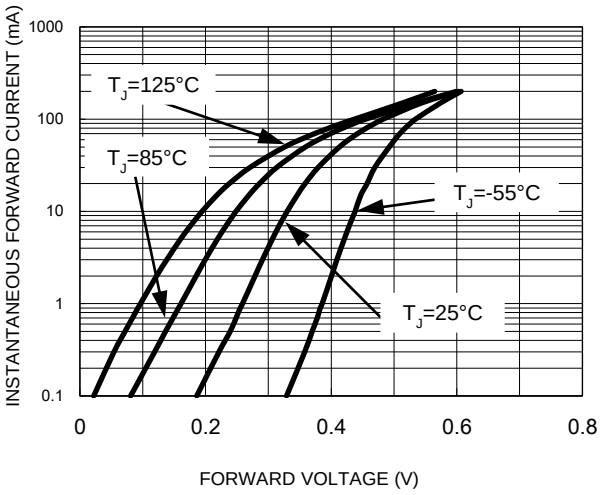
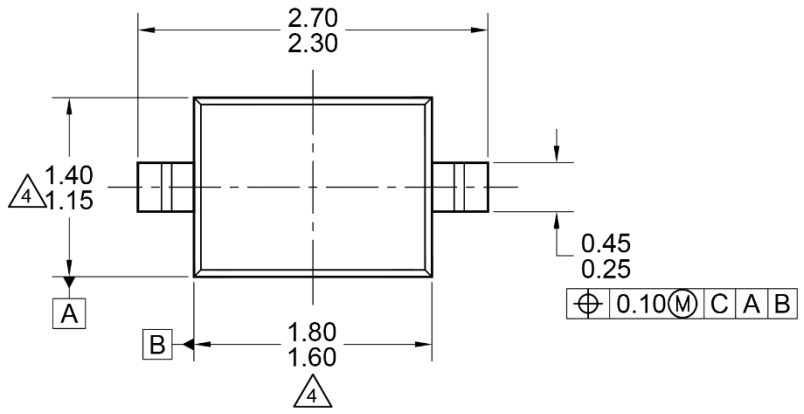


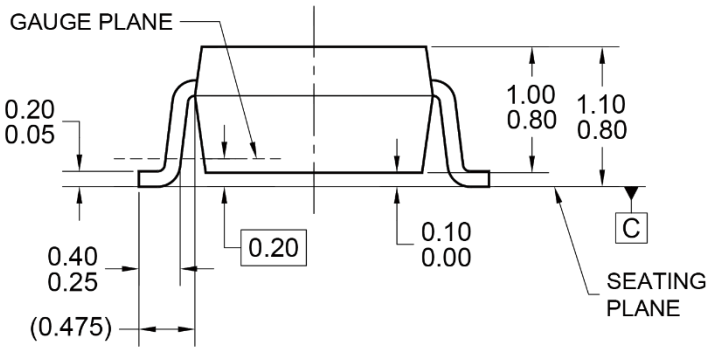
Fig.4 Typical Forward Characteristics

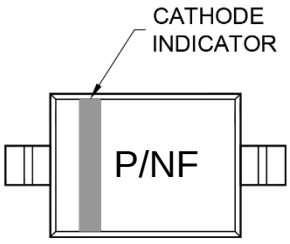


PACKAGE OUTLINE DIMENSION

SOD-323

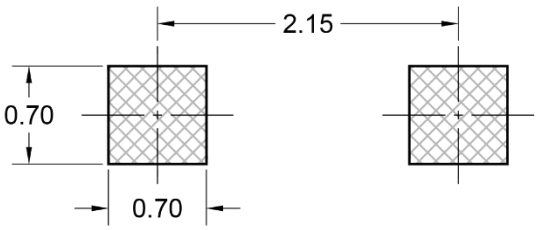






MARKING DIAGRAM

P/N = Device marking
F = Factory code



SUGGESTED PAD LAYOUT

NOTES: UNLESS OTHERWISE SPECIFIED

1. ALL DIMENSIONS ARE IN MILLIMETERS.
2. DIMENSIONING AND TOLERANCING PER ASME Y14.5M-1994.
3. PACKAGE OUTLINE REFERENCE: EIAJ ED-7500A, SC-76A.
4. MOLDED PLASTIC BODY LATERAL DIMENSIONS DO NOT INCLUDE MOLD FLASH, PROTRUSIONS OR GATE BURRS.
5. DWG NO. REF: HQ2SD07-SOD323-113 REV A.

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